

Introduction to Interactive Multimedia Applications

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Interactive

Multimedia

Applications

Interactive

Multimedia

Applications (hands-on part...)

Human-Centered Approach

A system implementor cannot pretend to be a user

- You know too much
- You can't forget

➔ Human-Computer Interaction (HCI) Mantra:
“The user is not like me!”

Human-Centered Approach

User/Human-Centered...

... Computing (HCC)

... Design (HCD or UCD)

(to contrast it with a focus on the technology itself)

Human-Centered Approach

Norm: ISO 9241 Ergonomics of human–system interaction

Part 210: Human-centered design for interactive systems

“Using a human-centered approach to design and development has substantial **economic and social benefits for users, employers and suppliers**. Highly usable systems and products tend to be more **successful both technically and commercially**.“

Human-Centered Approach

Norm: ISO 9241 Ergonomics of human–system interaction

Part 210: Human-centered design for interactive systems

- a) the design is based upon an explicit understanding of users, tasks and environments;
- b) users are involved throughout design and development;
- c) the design is driven and refined by user-centred evaluation;
- d) the process is iterative;
- e) the design addresses the whole user experience;
- f) the design team includes multidisciplinary skills and perspectives.

Human-Centered Approach

Norm: ISO 9241 Ergonomics of human–system interaction

Part 210: Human-centered design for interactive systems

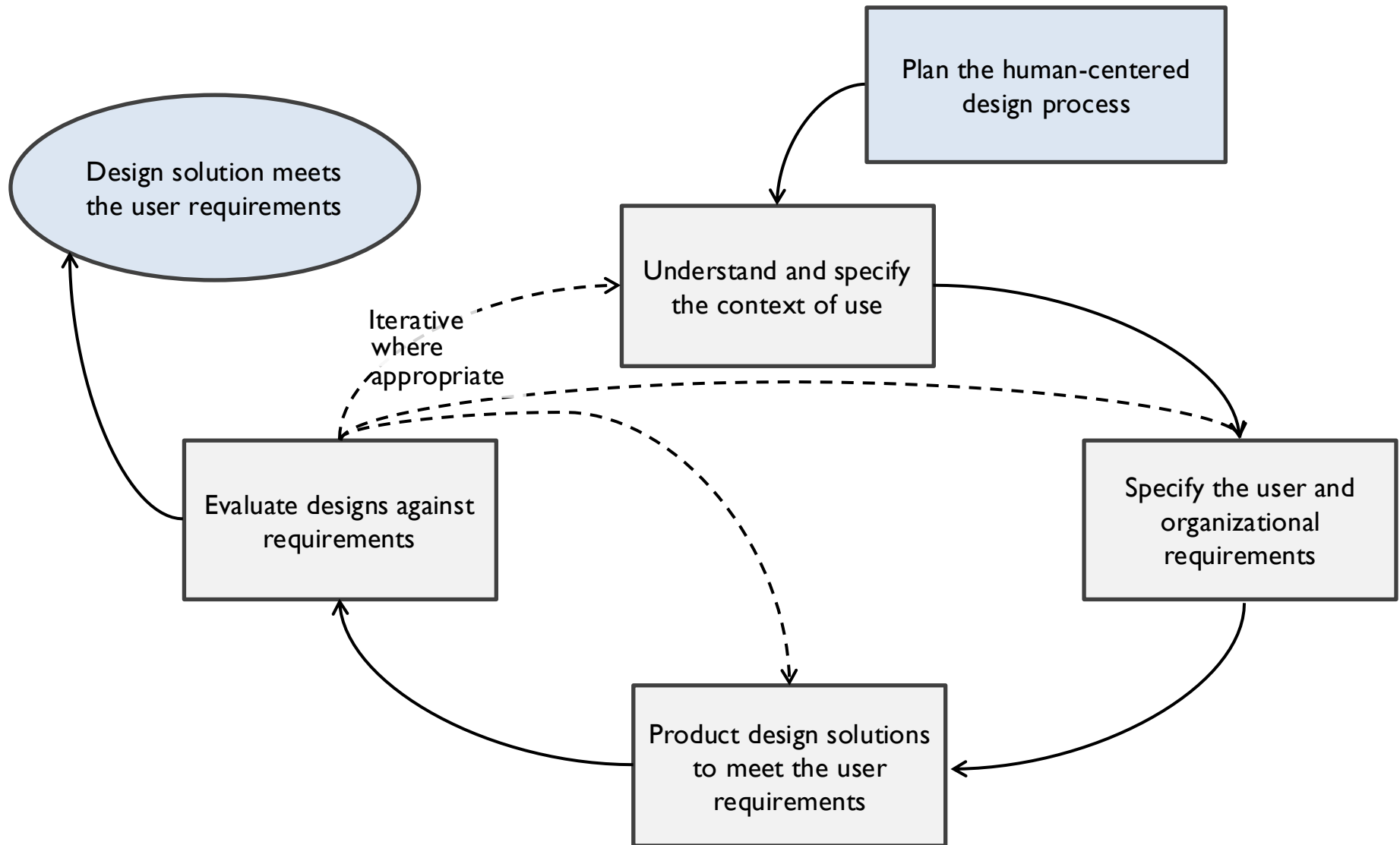
“a) understanding and specifying the **context of use**”

“b) specifying the **user requirements**”

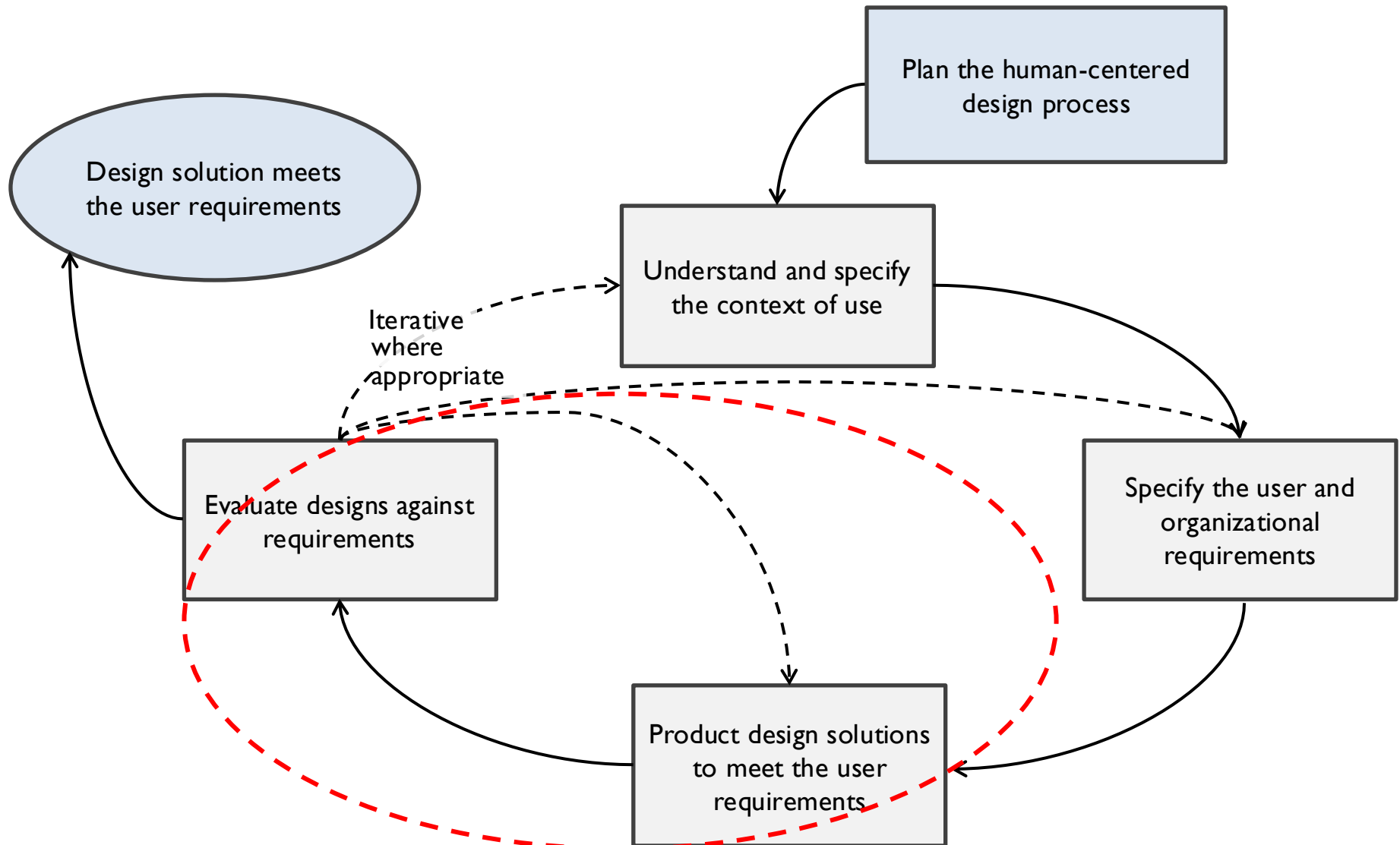
“c) producing **design solutions**”

“d) **evaluating** the design”

Human-Centered Approach



Human-Centered Approach



Human-Centered Approach

1st – Concept development and Interaction design
(quick iterations)

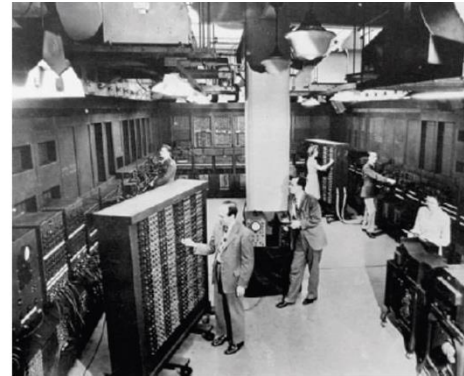
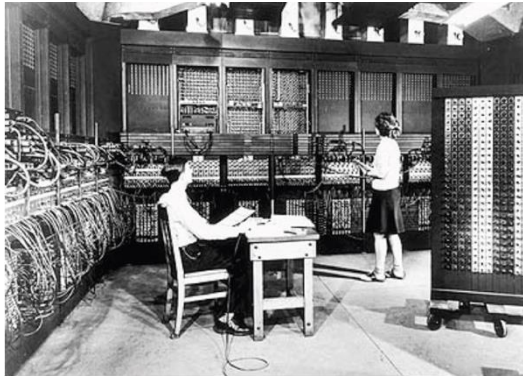
- Application and interaction concept
- Interaction design
- Prototypes to evaluate the concept and interaction design

2nd – technical realization (slow iterations)

- Technical analysis
- Technical specification (e.g. architecture, platform)
- Implementation
- Evaluation and Quality management

1940s/50s – First Computers

Electrical (first computers)



<https://youtu.be/bGk9W65vXNA?t=1m9s>

History of Computers

Memex: Vannevar Bush, *As We May Think*, 1945 (Hypermedia)

“Machines with interchangeable parts can now be constructed with great economy of effort. In spite of much complexity, they perform reliably. [...]The world has arrived at an age of cheap complex devices of great reliability; and something is bound to come of it.”

“Consider a future device for individual use, which is a sort of mechanized private file and library. It needs a name, and, to coin one at random, "memex" will do. **A memex is a device in which an individual stores all his books, records, and communications, and which is mechanized so that it may be consulted with exceeding speed and flexibility. It is an enlarged intimate supplement to his memory.**”

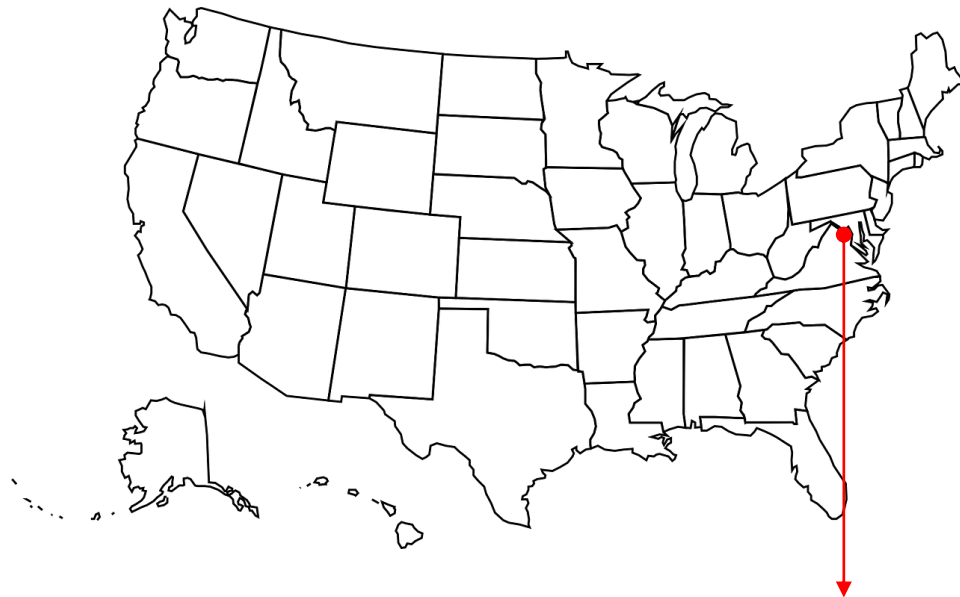
1950s – Interactive Computing



1960 – Interactive Computing



1960s – Interactive Computing

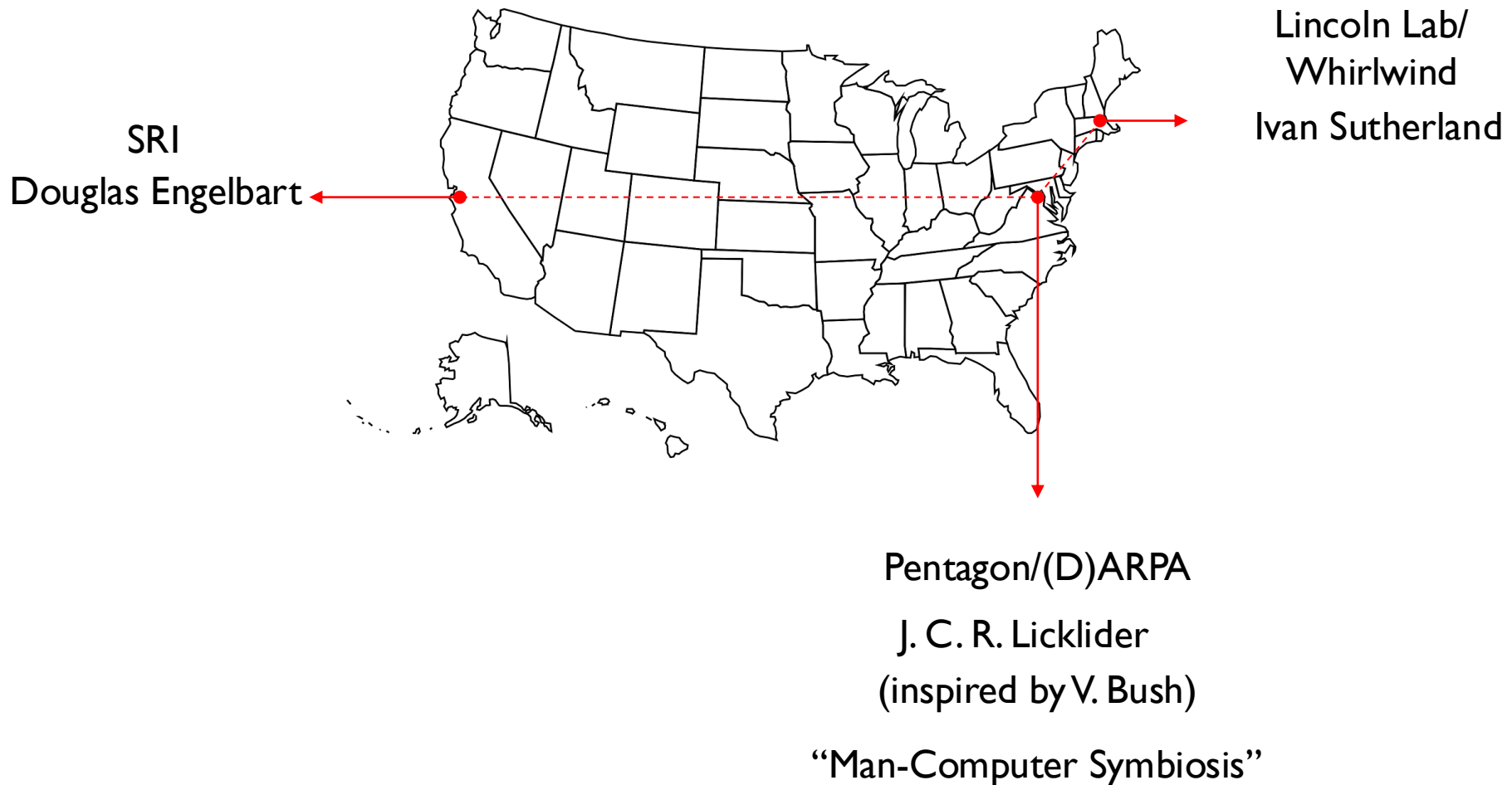


Pentagon/(D)ARPA

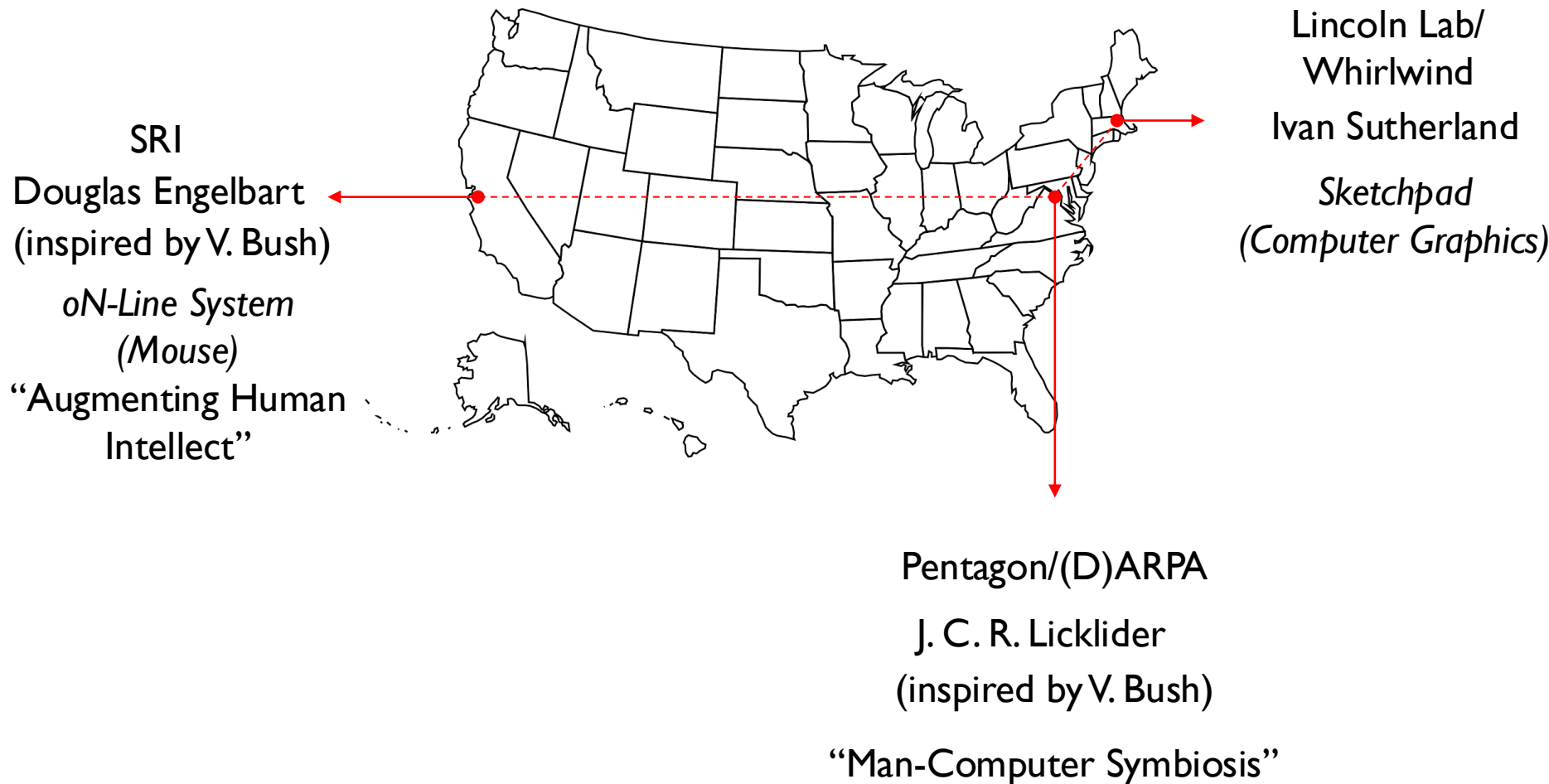
J. C. R. Licklider
(inspired by V. Bush)

“Man-Computer Symbiosis”

1960s – Interactive Computing



1960s – Interactive Computing



”The Mother of All Demos”

Augmenting Human Intellect: A conceptual framework. Douglas Engelbart, 1962. (NLS, oN-Line System)



<http://dougengelbart.org/firsts/interactive-computing.html>

<http://youtu.be/yJDv-zdhzMY?t=31m18s>

1960s – Interactive Computing

Whirlwind (1944-1950s/1960s) - TX-0 and TX-2:

- US Navy, US Air Force and MIT.
- Graphics on CRT and light gun/light pen
- Time-Sharing allowed several of simultaneous users working at terminals.

<http://youtu.be/06drBN8nlWg?t=16m1s>



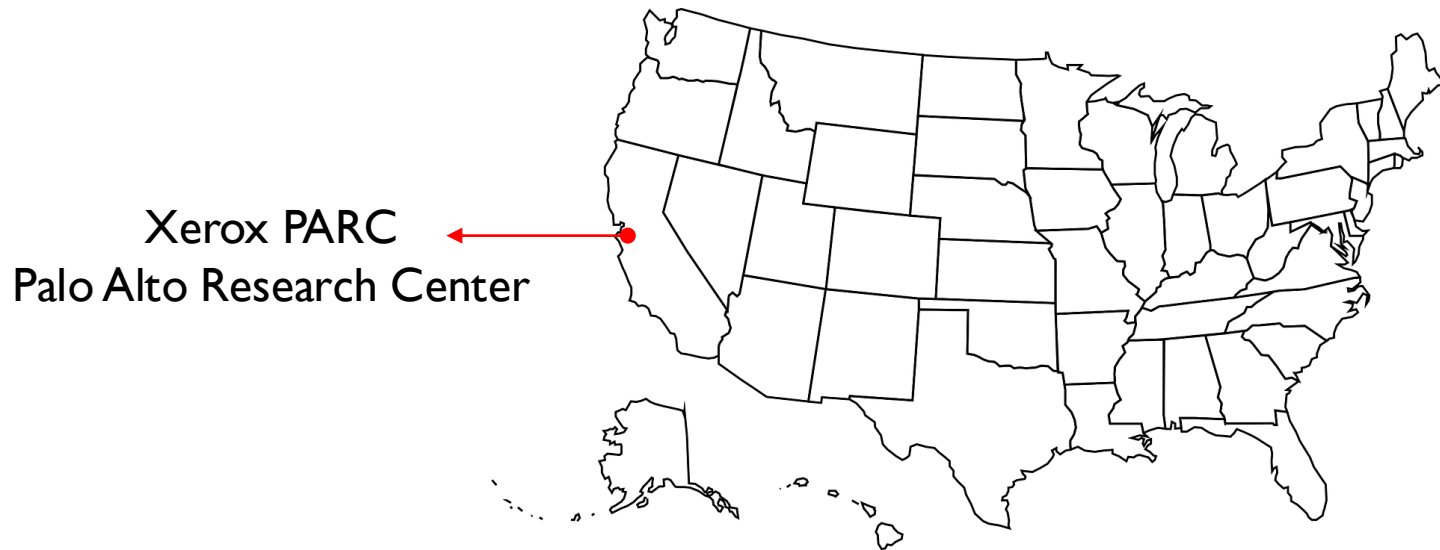
1960s – Interactive Computing

Sketchpad: Ivan Sutherland, PhD Thesis, 1963:

Most influential document in the history of HCI. His Sketchpad system, built on the TX-2 to make computers more approachable, launched computer graphics.

https://youtu.be/6orsmFndx_o?t=211

1970s – Interactive Computing



1970s – Personal Computer

Xerox Alto, 1973 (Xerox Star, 1981 - commercialized):
a window-based Graphical User Interface (GUI, WIMP*), icons, folders, mouse (two-button), Ethernet networking, file servers, print servers and e-mail.

<http://youtu.be/Cn4vC80Pv6Q?t=2m47s>

<http://youtu.be/ODZBL80JPqw?t=2m32s>

* "windows, icons, menus, pointer"



HCI & Design

William Bowman's 1968 book *Graphic Communication* was a strong influence on the Xerox Star, for which the designer Norm Cox's icons were chosen (Bewley et al., 1983). However, graphic design was considered a secondary activity (Evenson, 2005).

In 1990s, CHI embraces Design: **Design of Everyday Things**, Donald Norman, 1990; **Bringing Design to Software**, Terry Winograd, 1996.

Design brings Participatory Design (1970s) to HCI: involving specific people who would interact with a resulting system

Brain Drain from Xerox PARC

1980s – Xerox decide to focus their development on laser copy machines (a real investment for company, meaning profit...)

Consequence: Top researchers moved out of Xerox PARC and went to other Silicon Valley companies.... Including Apple (e.g., Larry Tesler, Alan Kay)!

From Command Line like Apple II (and MS-DOS)

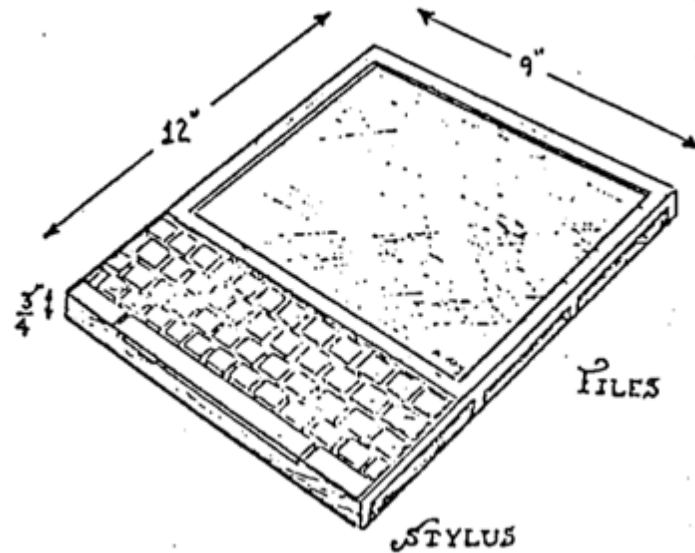
<https://www.youtube.com/watch?v=CxJwy8NsXF8>

To GUI like Apple Lisa! (and Windows 3.1)

<https://www.youtube.com/watch?v=OKrvMStjblU>

Mobile Computing

In 1973, Alan Kay describes the Dynabook, in *A Personal Computer for Children of All Ages*. (Computers start to be **Mobile**)



Mobile Computing

In 1973, IBM develops **SCAMP**, the first portable computer prototype and, in 1975 launches the commercial version IBM 5100. Xerox NoteTaker in 1978, followed by the commercial version Osborne 1 (1981). In 1982, Bill Moggridge (IDEO) designed the **Grid Compass (1101)**. Following years by EPSON HX-20; Gavilan SC and GridCase 1 (with batteries).

IBM
5100



Grid
Compass



Picture: <http://www.oldcomputers.net/>

https://youtu.be/DAHSS_Vgfl

Mobile Computing

In 1984, PSION lunches the first PDA, called **Organizer**. Around 1986, lunches the **CASIO IF-8000** (first calculator and organizer with stylus). In 1992, Apple presents the Apple Newton and IBM presents **IBM Simon**, the first smartphone. Following years: first tablets (GridPad and GO Tablet) and more PDAs/smarphones (Nokia 9000 and Palm).

PSION
Organizer



CASIO
IF-8000



Picture: Bill Buxton Collection

IBM
Simon



Picture: Bill Buxton Collection

Go Tablet and PenpointOS

GO Tablet: first tablet with operating system (PenpointOS) designed specifically for pen-based interactions.

<https://www.youtube.com/watch?v=x0XE08BjQDQ>

Mobile Computing

1995 - First mobile phone
with GPS (NAVSYS) -
Emergency Vehicle
Location System



<http://gpsworld.com/first-gps-cell-phone-on-display-at-smithsonian/>

1999 – Benefon ESC!
one of the first cellular
devices with a built in GPS



Mobile Computing

2000 – J-Phone First mobile phone with camera



<https://en.wikipedia.org/wiki/J-SH04>

2005 – iPod (5th Generation)
First that plays audio, images and videos



“Computers” are....

Small, cheap, powerful...



... and
everywhere



What is Multimedia?

Different viewpoints of the term “**multimedia**”: Consumer entertainment vendor? Hardware vendor? Designer? Computer scientist?

Multi: many.

Media: plural of *medium*.

Medium: storage and/or transmission tools used to store and deliver information or data.

Storage/Transmission: Input/Output -> Interactivity and Communication

Data: text, audio, images, drawings, animation, video. (GPS, physiological data)

Human Communication

Five senses:

Sight (vision): writing, images, moving images, body gestures

Hearing (auditory): speech, sound/music

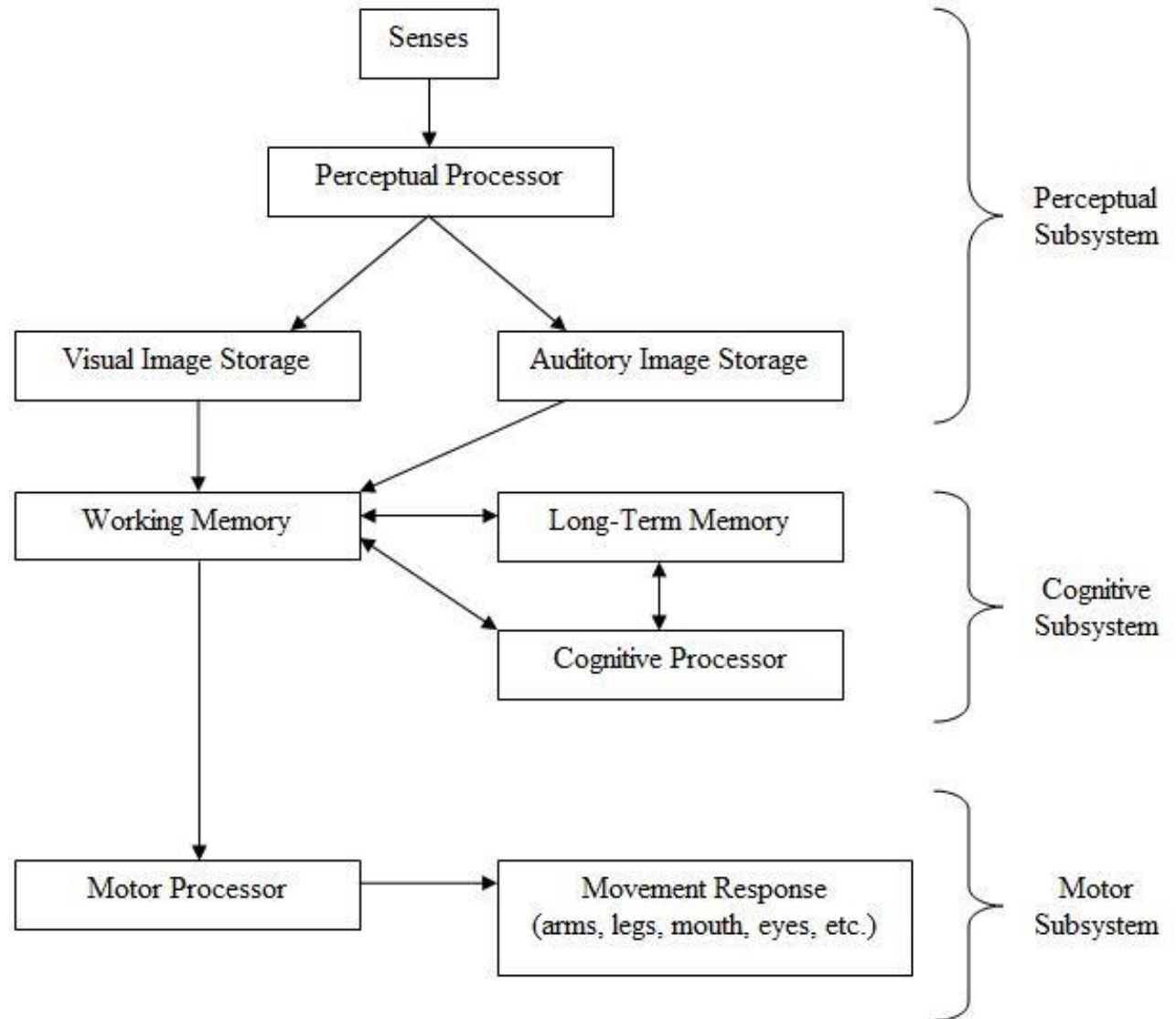
Touch and proprioception (somatosensory)

Smell (olfactory)

Taste (gustatory)

The Model Human Processor

Stuart K. Card,
Allen Newell, and
Thomas P. Moran.
1983.
*The Psychology
of Human-Computer
Interaction.*
L. Erlbaum
Assoc. Inc., Hillsdale, NJ
USA.



Perception

The process of gathering **awareness** or **understanding** of surroundings by **interpreting sensory information**

Perception

Mechanosensitivity: Examples of mechanosensitivity are kinaesthesia (sense of own movement), touch, hearing, and equilibrium. Here, the energy of physical contact is transduced, for example, through hair cells in the ear.

Chemosensitivity: Examples of chemosensitivity include gustation and olfaction. Here, chemical properties are transduced, for example, by taste buds on the tongue.

Photosensitivity: Examples include the retina. Here, the stimulation by photons is transduced into electrochemical events in the rod and cone cells.

Awareness and Perception

The users's perception of the surroundings is based on human senses, but relates at the same time to experience and memory.

Five senses:

Sight (vision)

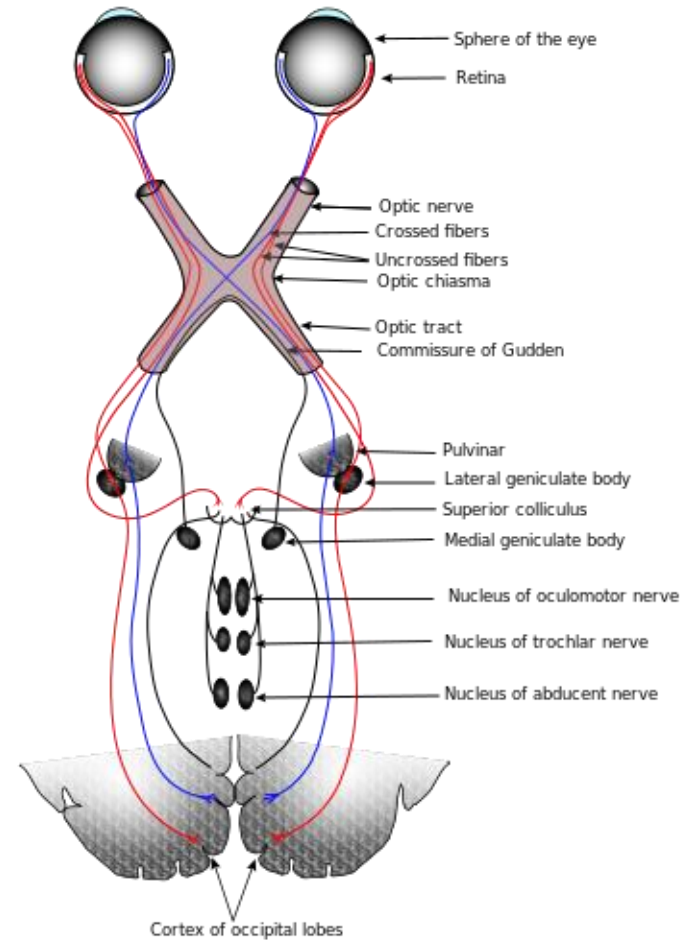
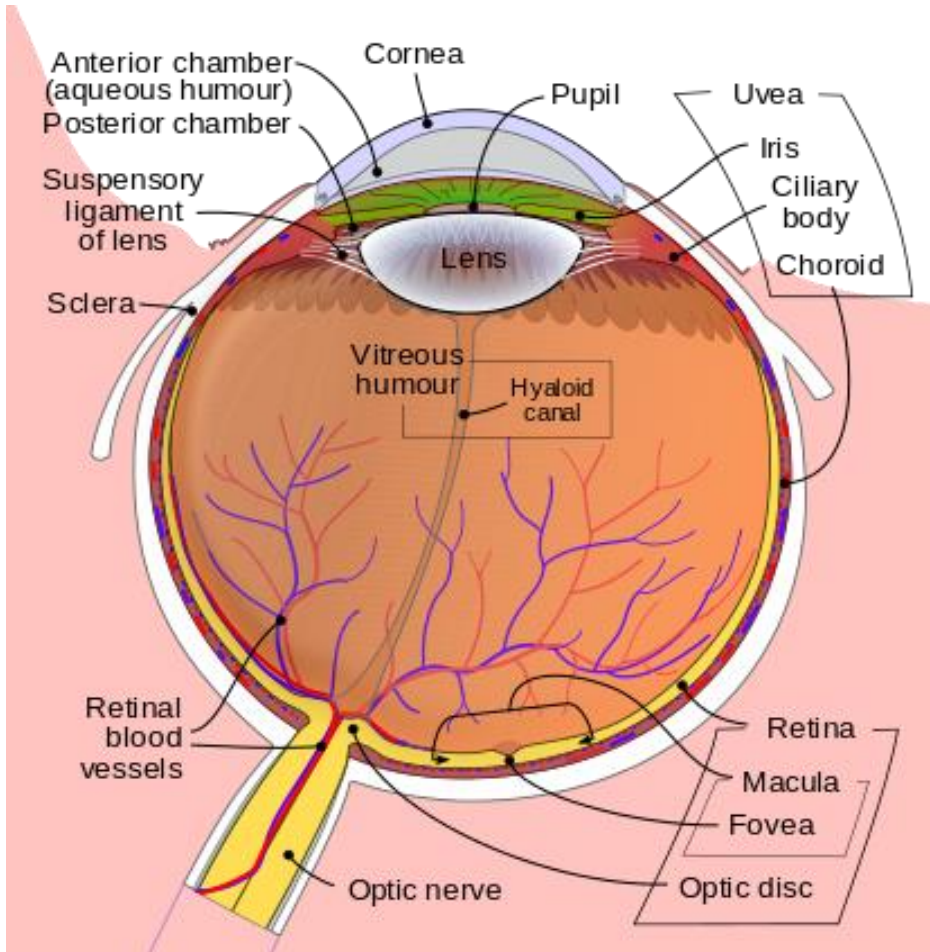
Hearing (auditory)

Touch and proprioception (somatosensory)

Smell (olfactory)

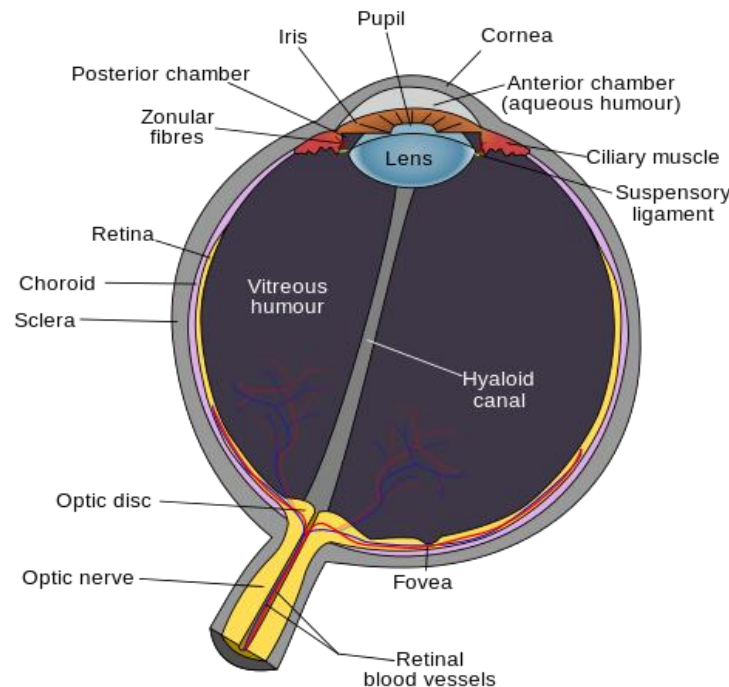
Taste (gustatory)

Vision



Human Vision - Retina

Photoreceptor cells: cones (color: red+green+blue); rods (grayscale)
and photosensitive retinal ganglion (light information to the body)
Sends light information to visual cortex (brain) for processing.



What is Color

Color is determined by the wavelength
(or equivalently frequency) of light

Eye can sense

~380-750nm (nano-meters: wavelength) ~789-400THz (tera-hertz: frequency)

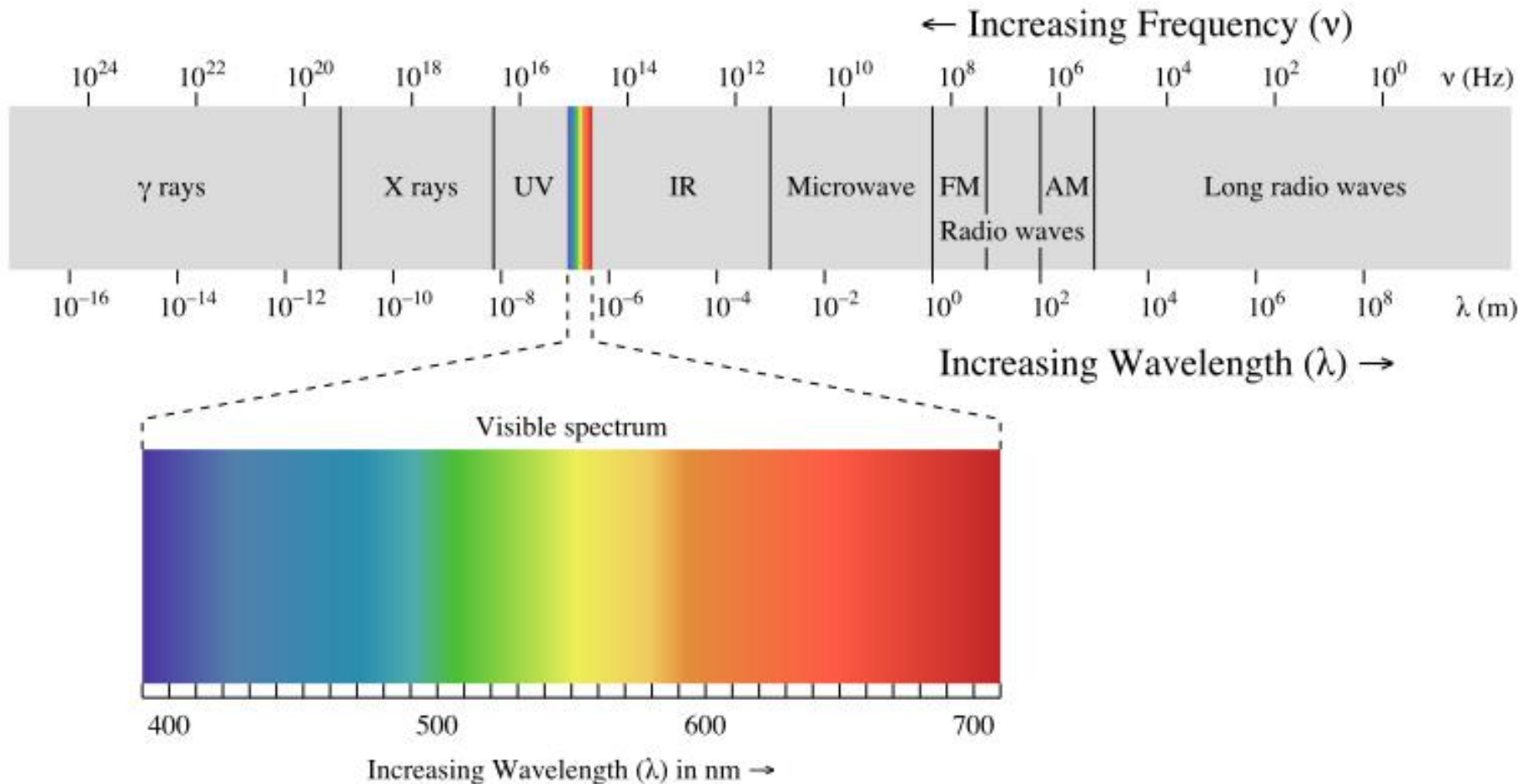
violet → blue → cyan → green → yellow → orange → red

~380-450nm ~450-475nm ~475-495nm ~495-570nm ~570-590nm ~590-620nm
~620-750nm

Light can be a mixture of frequencies

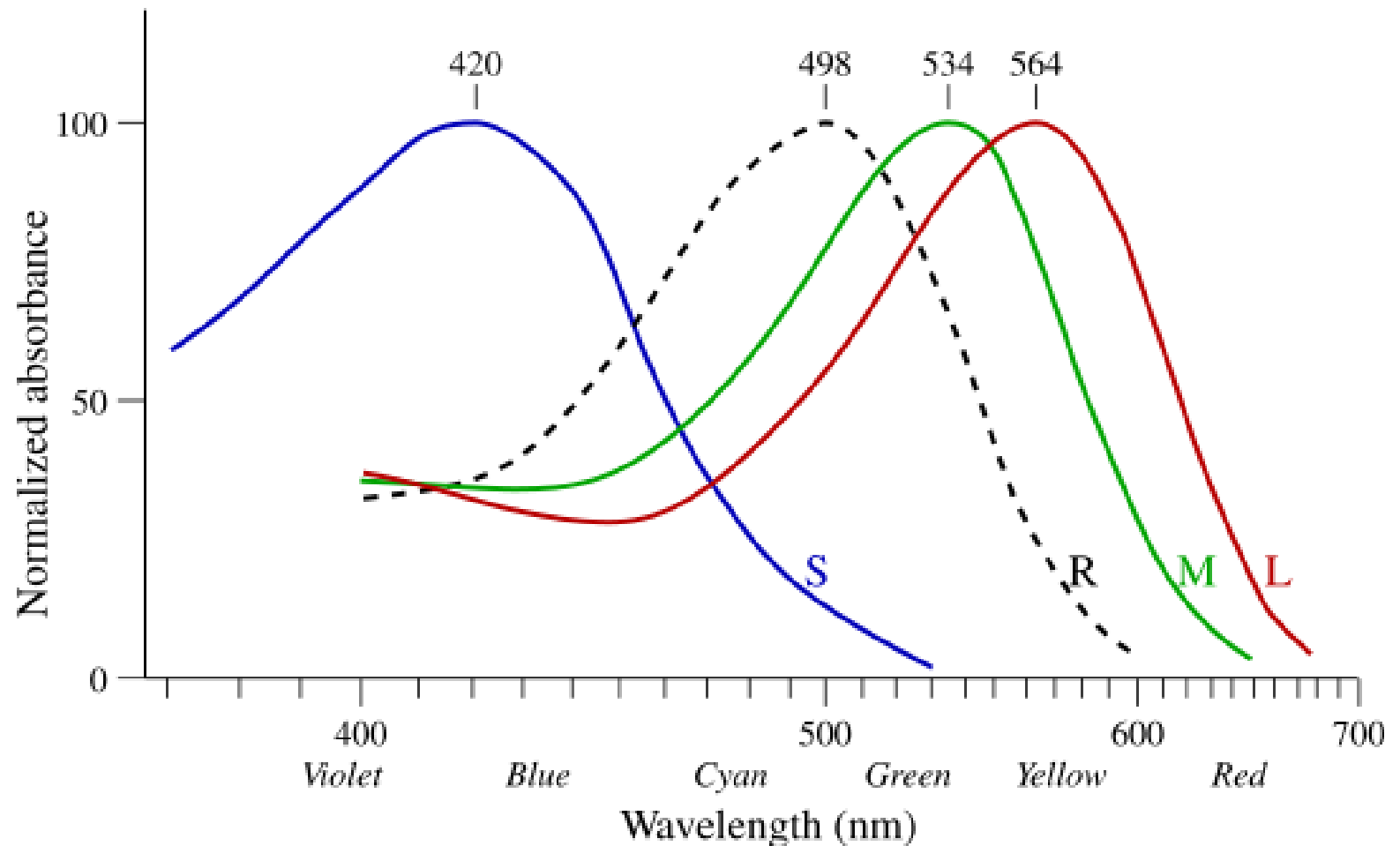
so ***color is really a distribution of frequencies***

Human Vision – Visible Waves



Human Vision

Eye naturally samples



Human Vision

Frequency range 430 – 790 THz (390 – 700 nm)

Visible light

Foveal vs. peripheral vision

Resolution varies on the retina

Fovea is 1% of the area but takes 50% of visual cortex

Field of view (including peripheral vision)

Up to 180° horizontal without moving the eyes

Up to 270° horizontal moving the eyes

Stereo (binocular) vision (3D/Depth information)

Human Vision - Color

Color perception usually involves three quantities:

Hue: Distinguishes between colors like red, green, blue, etc

Saturation: How far the color is from a gray of equal intensity

Lightness: The perceived intensity of a reflecting object

Sometimes lightness is called brightness if the object is emitting light instead of reflecting it.

<https://www.youtube.com/watch?v=2AEaATI9DY4>

Human Vision - Depth

Monocular cues

Motion parallax, depth from motion, occlusion, perspective...

Binocular cues

Stereopsis, convergence... (3D: same image from 2 different angles)

Human Vision - Motion

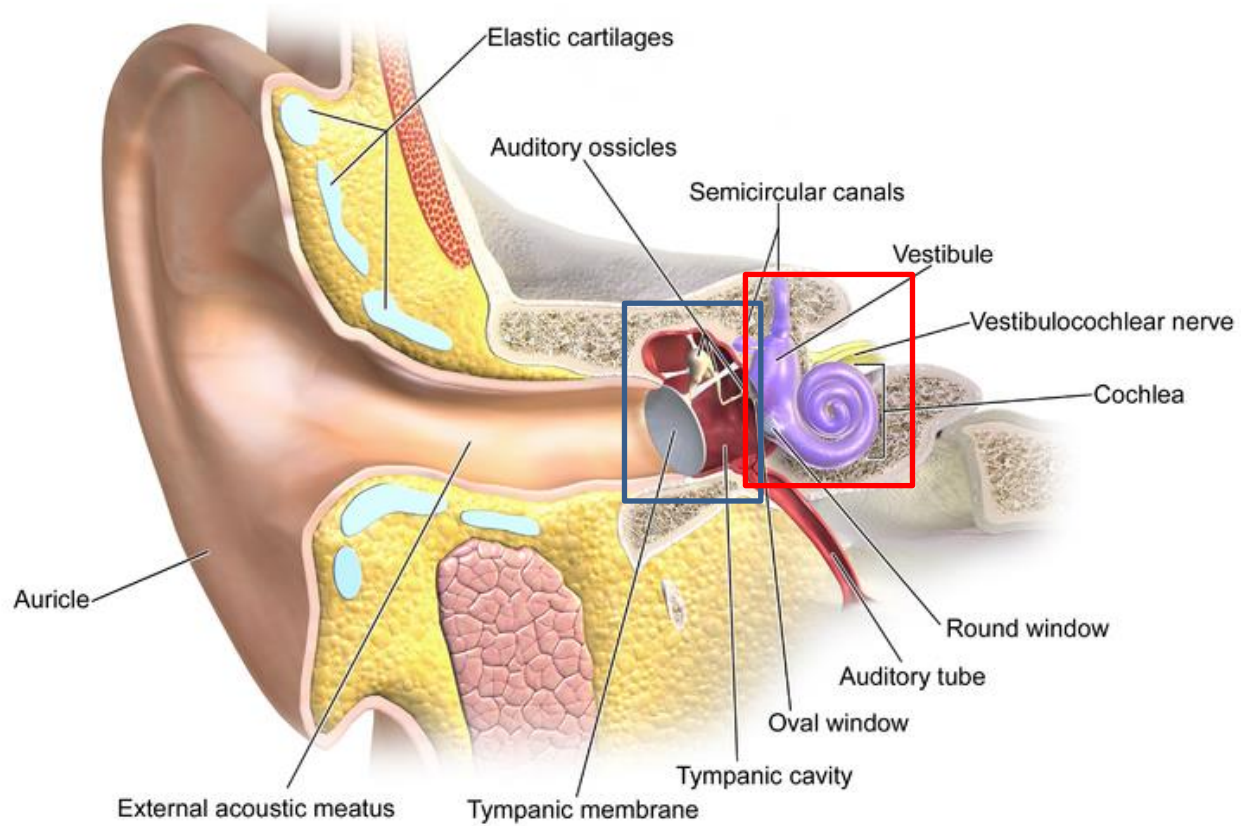
Motion perception

First order: by luminance changes (Phi and Beta Phenomenons)

Frame rate is greater than 10 to 12 separate images per second.

Second order: contrast and texture.

Auditory

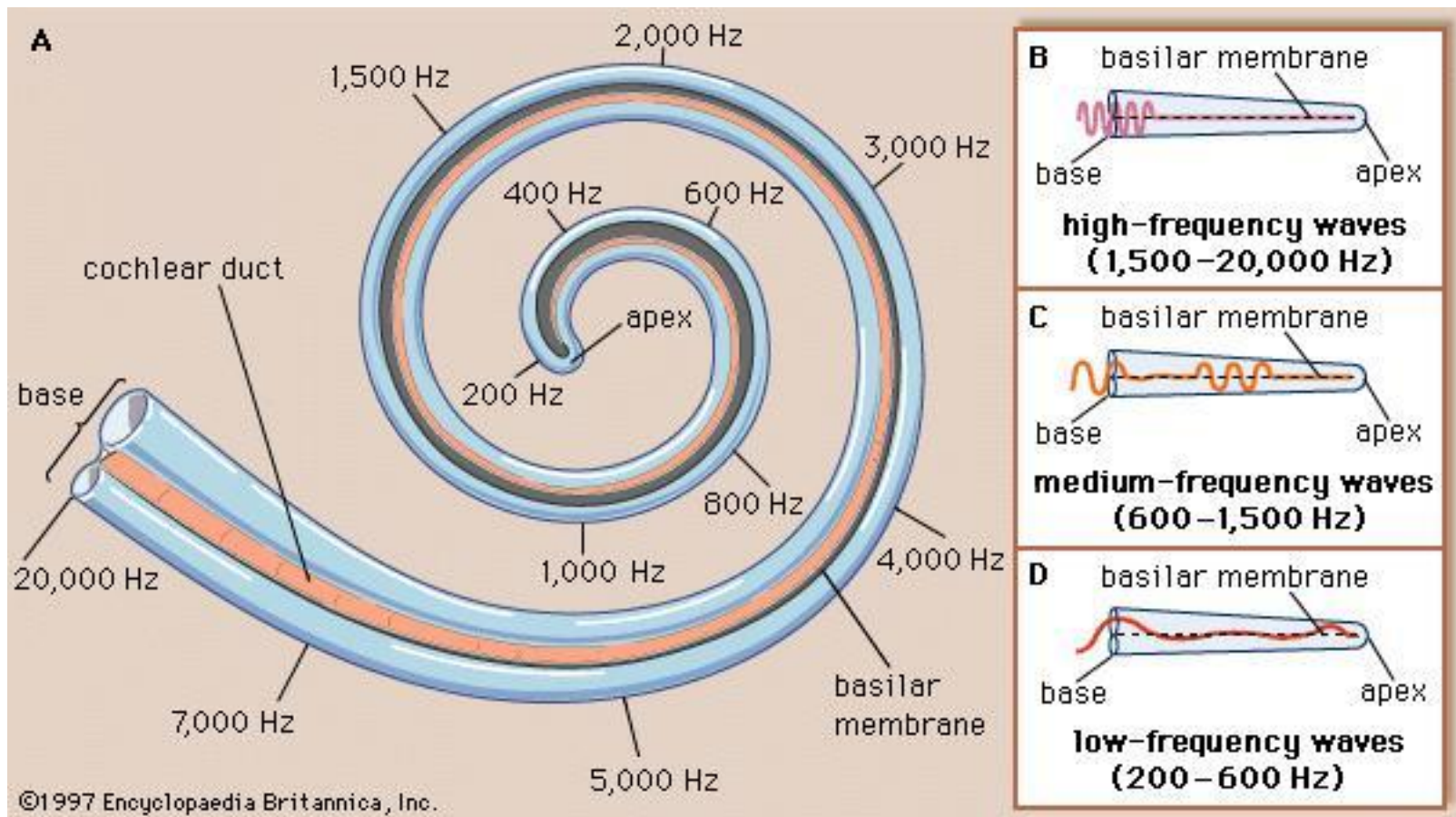


The Anatomy of the Ear

Human Hearing

Sound waves travel through the outer ear are modulated by the middle ear, and are transmitted to the vestibulocochlear nerve in the inner ear. This nerve transmits information to the temporal lobe of the brain, where it is registered as sound.

Human Hearing - Cochlea



Human Hearing

Frequency range: 20 Hz – 20 000 Hz

Nearly 10 octaves

In practice, the higher end is usually lower and decreases with age

Audibility depends on loudness

- We can perceive

Pitch

Loudness

Direction

Timbre

Duration

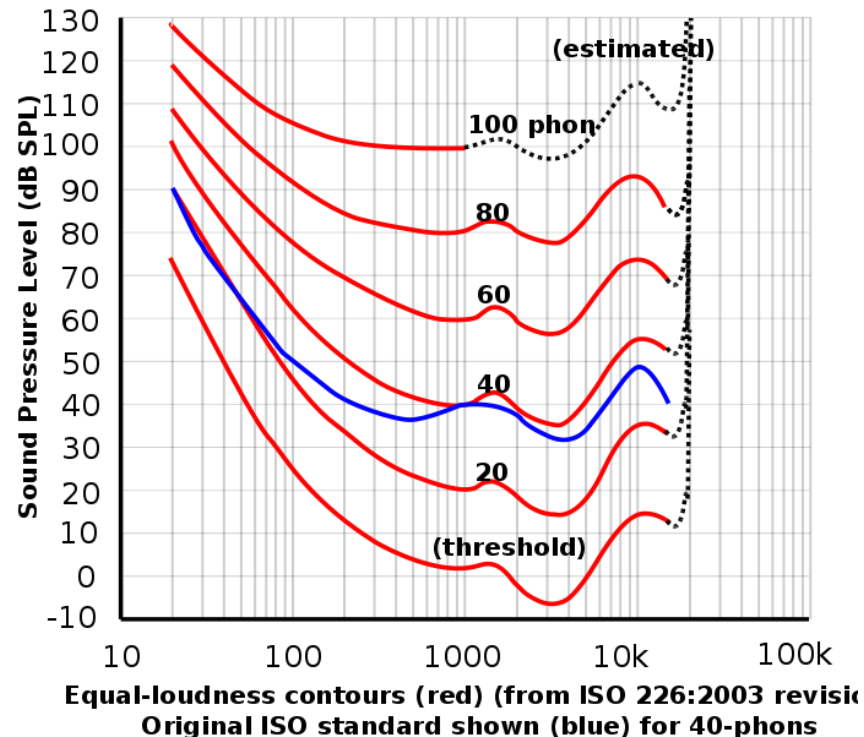
Rhythms

Human Hearing

Frequency separation is frequency dependent

Better frequency separation at low frequencies than high freqs

- Sensitivity/loudness perception is frequency dependent



Human Hearing

We have two ears

Binaural hearing is the basis of localizing sound sources (although there are also monaural cues)

Key factors depending on source location:

- ITD (interaural time difference)
- ILD (interaural level difference)

The **cocktail party effect** is the phenomenon of being able to focus one's auditory attention on a particular stimulus while filtering out a range of other stimuli, much the same way that a partygoer can focus on a single conversation in a noisy room.

Human Hearing

Auditory perception of distance:

Not as accurate as visual perception

- Theory: largely based on the ratio of direct sound and reverberant sound

Due to sound attenuation, the further we are from the source, the less direct sound there is to perceive

Human Visual and Audio Perception

	Visual	Auditory
Freq. range (Hz)	$(430 - 790) \cdot 10^{12}$ Hz	20 – 20000 Hz
Freq. range (octaves)	1	10
Directional	Yes (limited to field of view)	Yes (~ 360°)
Distance perception	Accurate	Limited
Size perception	Accurate	Limited
Rhythm perception	Limited	Accurate
Reaction time	Fast (~330 ms)	Faster (~280 ms)
“Color” perception	Millions of colors	Even more (timbre)

Proprioception and Somatosensory

Somatic – “of the body”

Relating to body

Pertaining to body wall in contrast to viscera

Distinguished from germplasm or psyche

Touch, pressure, vibration, limb position, heat, cold and pain

Proprioception (*kinesthetic sense*): the relative position of neighboring parts of the body and strength of effort being employed in movement. (“human physical output”)

Somatosensory System

A complex system of nerve cells that responds to changes to the surface or internal state of the body.

Nerve cells called "sensory receptors":

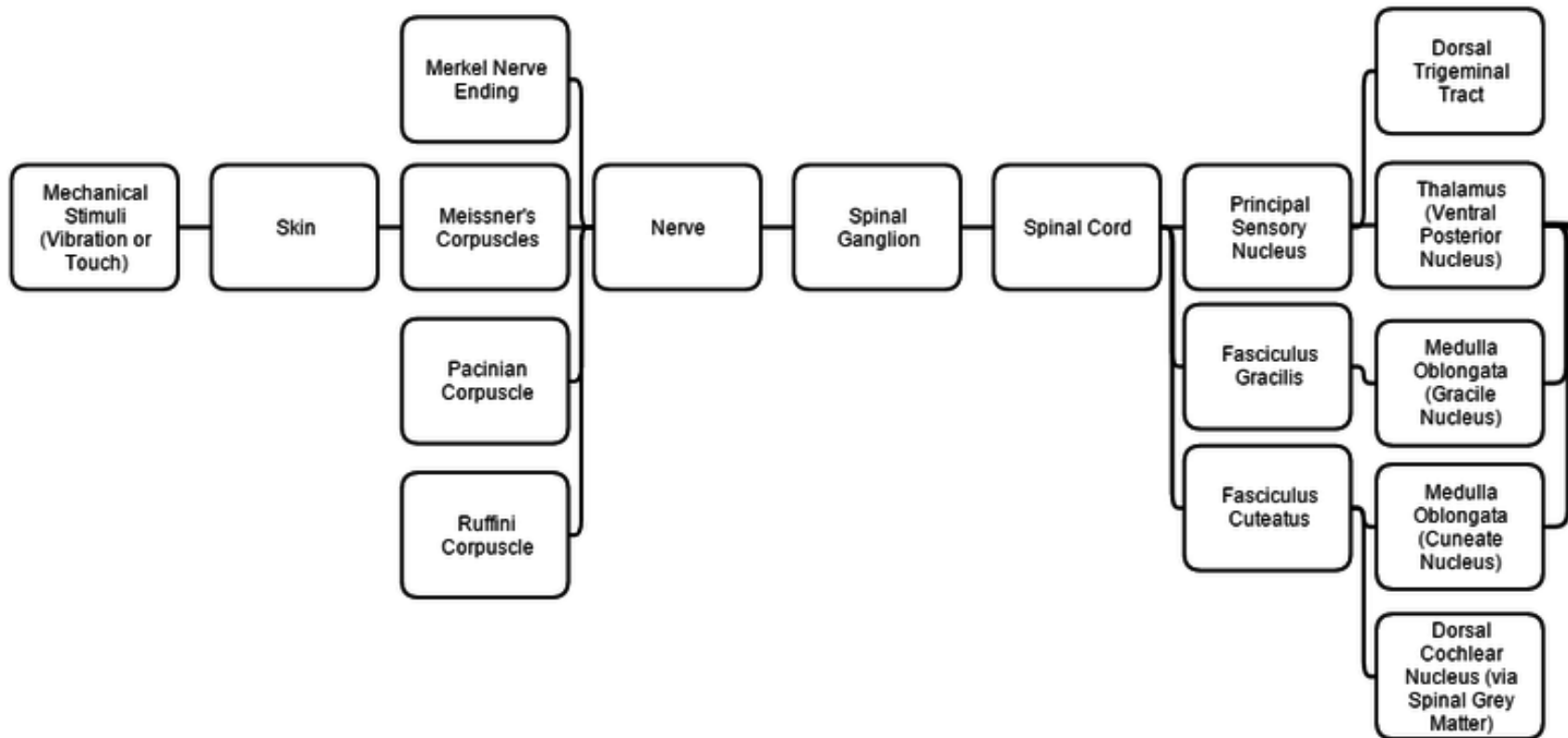
- Thermoreceptors: temperature

- Mechanoreceptors: pressure or distortion

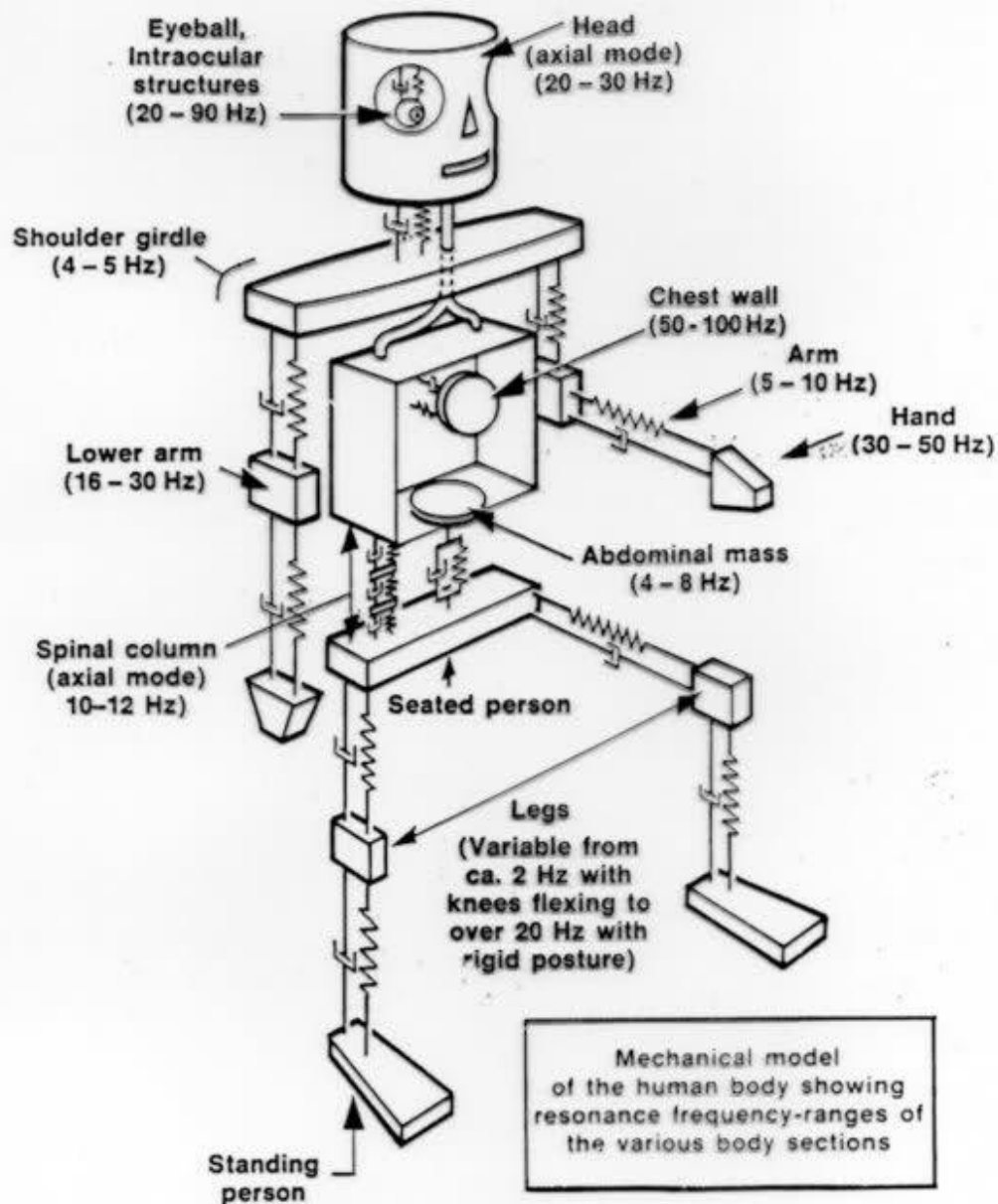
- Nociceptors: damage/pain

- Chemoreceptors: chemical signal

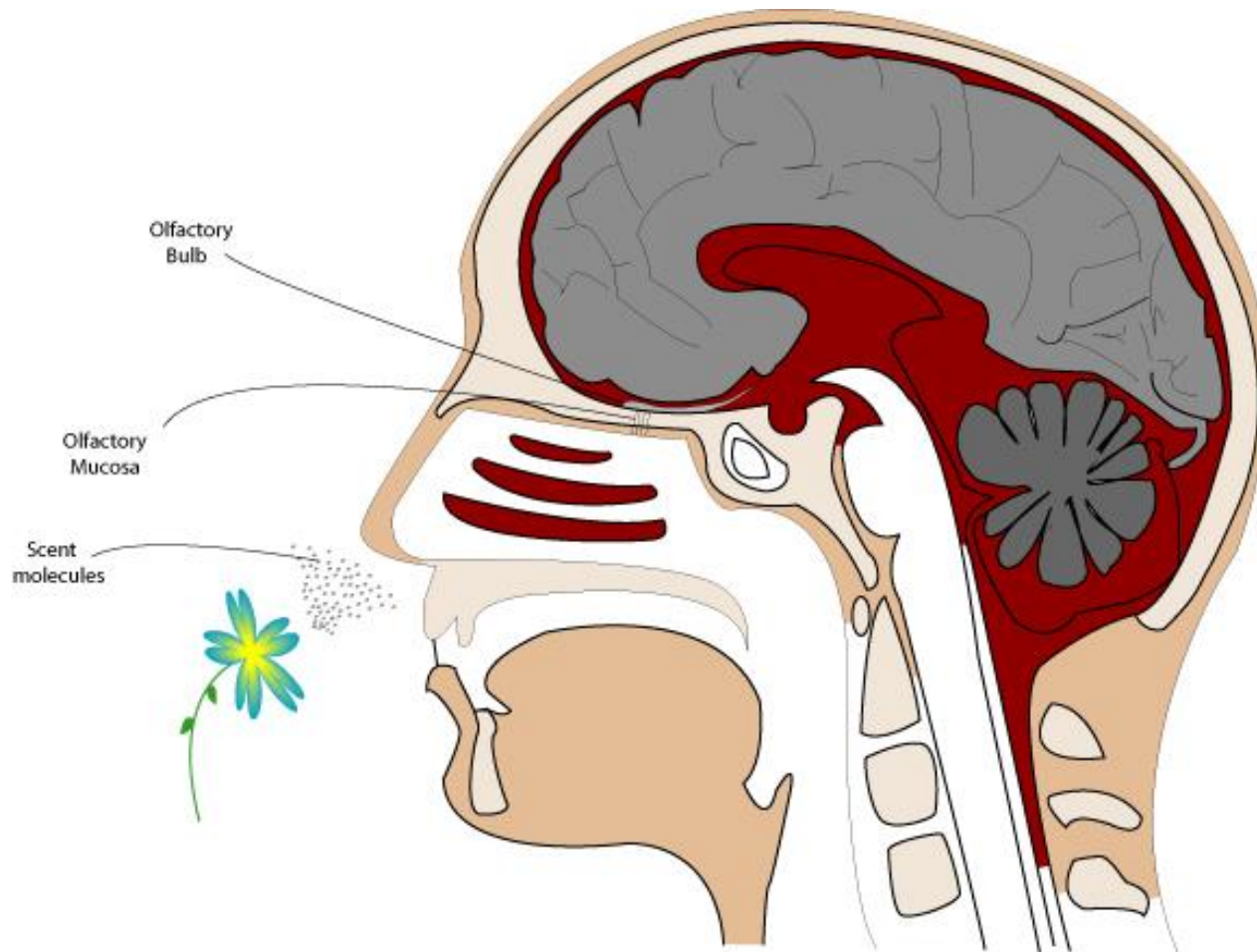
Somatosensory



Human body resonance frequencies



Olfactory



Gustatory

